Chapter 5: Multivariate Regression

Jonathan Roth

Mathematical Econometrics I
Brown University

Outline

1. Deriving Multivariate Regression and OLS

2. Regression and Causality

3. Regression Odds and Ends

Moving Beyond One "Regressor"

- So far we've talked about regression as a way of approximating the CEF $E[Y_i|X_i = x] \approx \alpha + x\beta$ for a single scalar X_i
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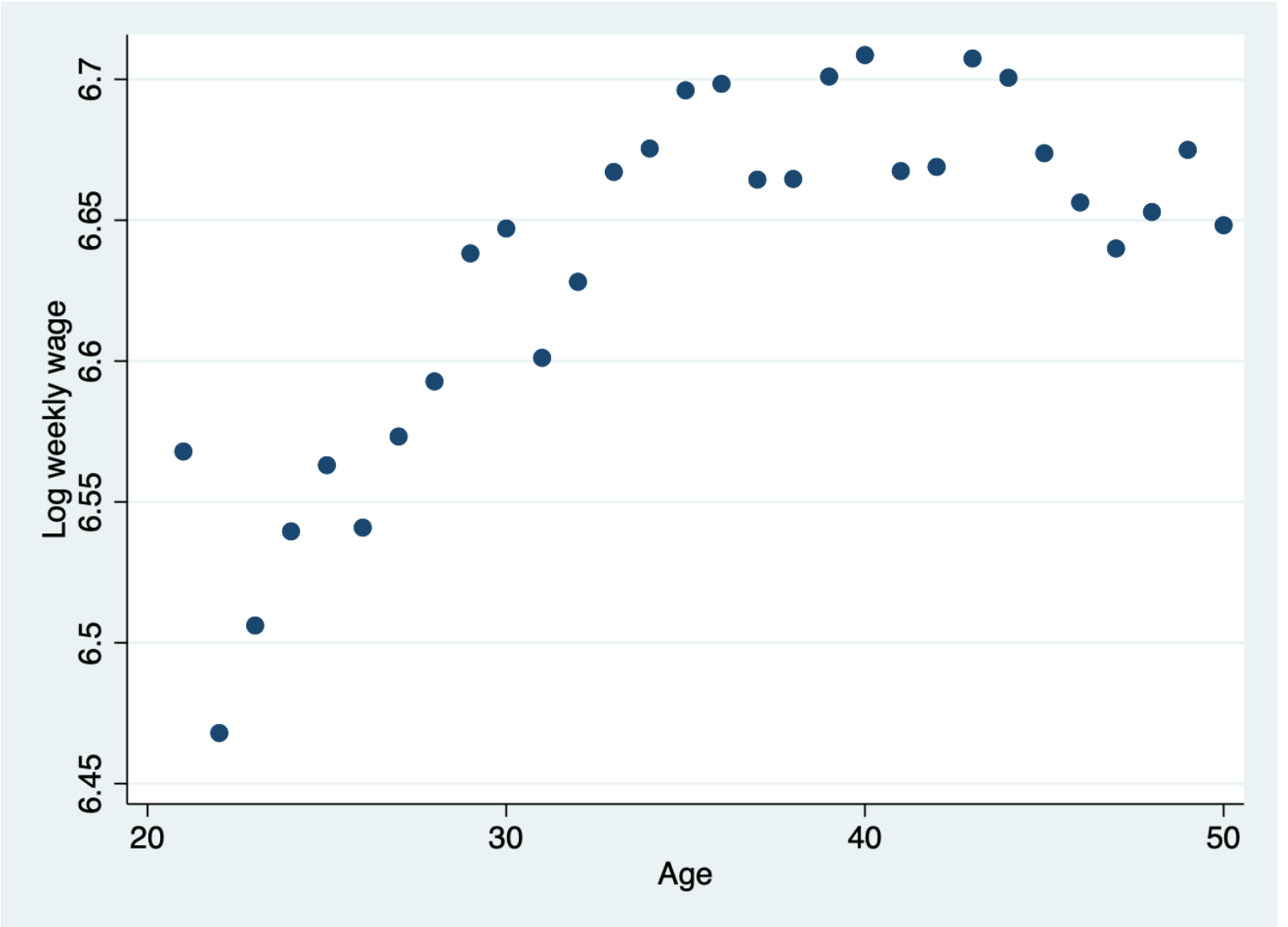
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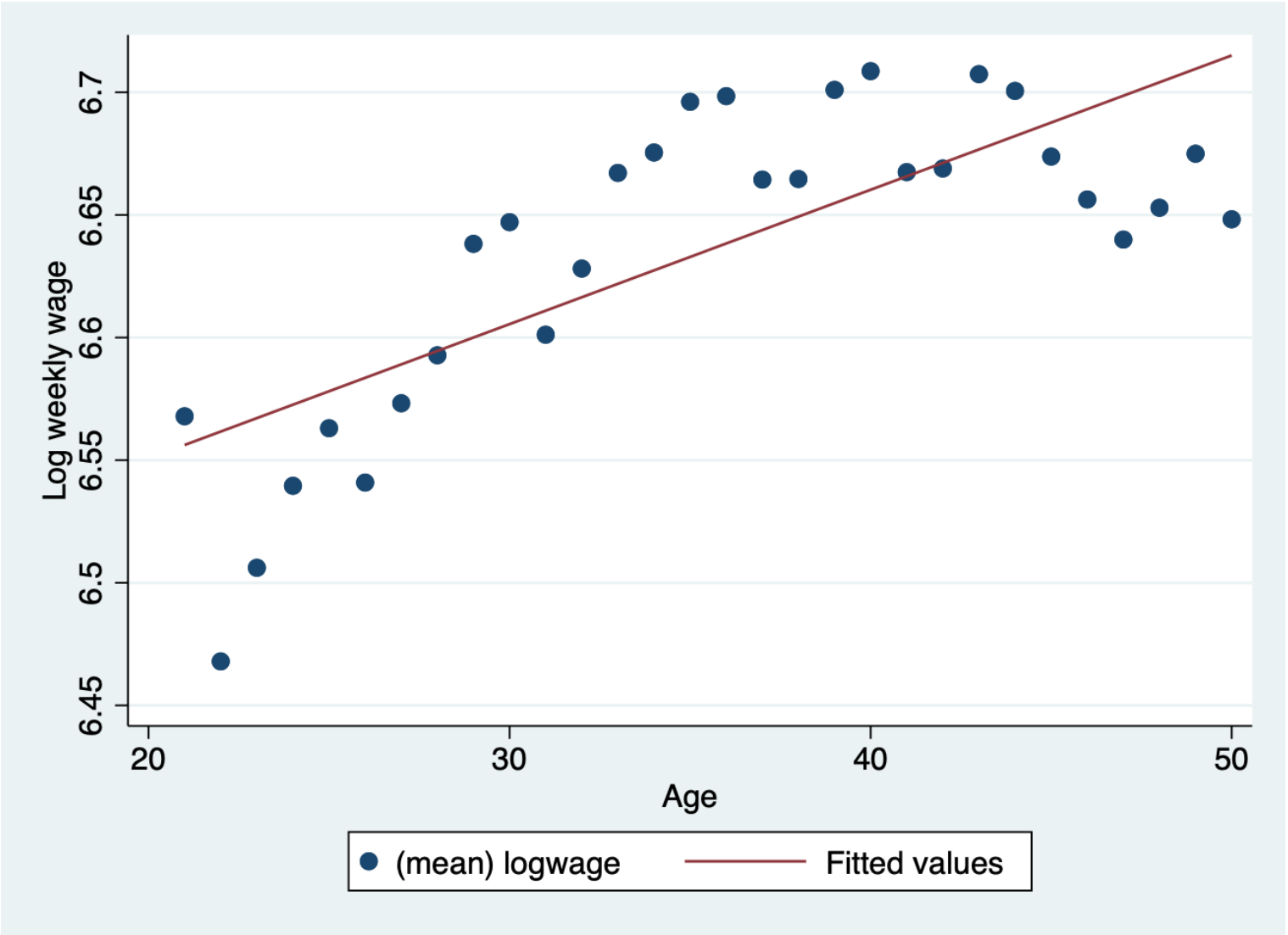
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   - In the Brown/URI example, we may want to control for high school GPA, family income, SAT, race ...
2. We want a *nonlinear* CEF approx.: e.g. $E[Y_i \mid X_i] \approx \alpha + X_i\beta + X_i^2\gamma$
   - We can "trick" regression into doing this by setting $\mathbf{X_i} = (1, X_i, X_i^2)'$

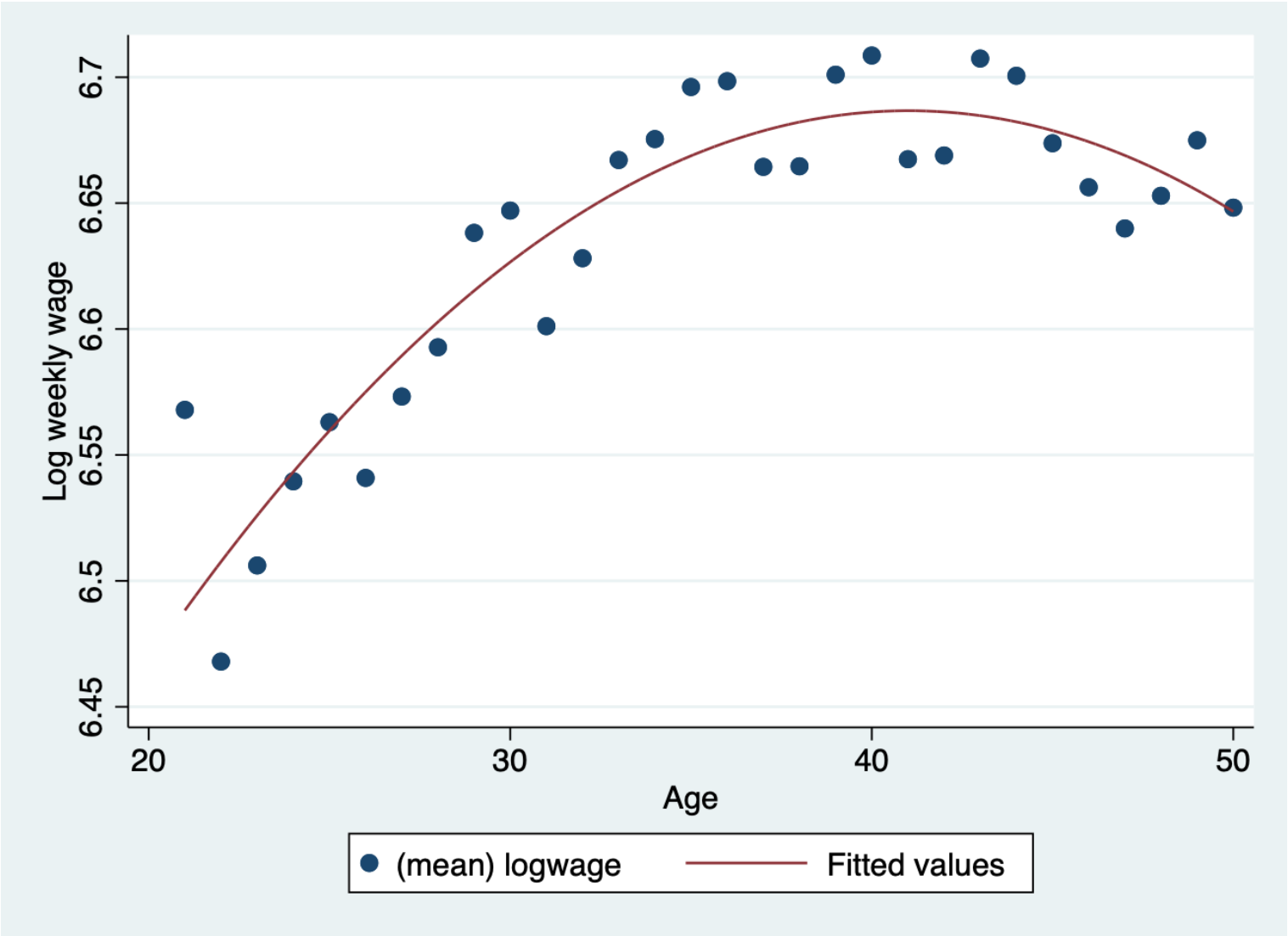
Log Wages by Age



OLS Regression (Linear Fit)



OLS Regression (Quadratic Fit)



Multivariate Regression as a Least-Squares Problem

- Recall with univariate OLS we solved for

$$(\alpha, \beta) = \arg\min_{a,b} E[(Y_i - (a + bX_i))^2]$$

We showed if the CEF is linear, then $E[Y|X] = \alpha + \beta X$; while if not, $\alpha + \beta X$ gave the best non-linear approximation

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- We will now consider the multi-variate analog:

$$\boldsymbol{\beta} = \arg\min_{\mathbf{b}} E\left[(Y_i - \mathbf{X}_i'\boldsymbol{b})^2\right]$$

- Using similar steps for the univariate case, we can show that if the CEF is linear in $\mathbf{X}$, then $E[Y|\mathbf{X}] = \mathbf{X}'\boldsymbol{\beta}$; if not, then $\mathbf{X}'\boldsymbol{\beta}$ is the MSE-minimizing approximation to the CEF.

Solving for Regression Coefficients

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\Rightarrow &E[\mathbf{X}_i Y_i] = E[\mathbf{X}_i\mathbf{X}_i']\boldsymbol{\beta}
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Multivariate Regression, in the Population and Sample

- Solving for $\boldsymbol{\beta}$, we obtain an expression involving population means:

$$\boldsymbol{\beta} = E[\mathbf{X}_i \mathbf{X}_i']^{-1} E[\mathbf{X}_i Y_i]$$

  - With a bit of algebra, you can show that this reduces to the bivariate formulas $\beta = \frac{Cov(X_i, Y_i)}{Var(X_i)}$ and $\alpha = E[Y_i] - E[X_i]\beta$ when $\mathbf{X}_i = (1, X_i)'$

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$$\hat{\boldsymbol{\beta}} = \left(\frac{1}{N} \sum_{i=1}^{N} \boldsymbol{X}_i \boldsymbol{X}_i' \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^{N} \boldsymbol{X}_i Y_i \right)$$

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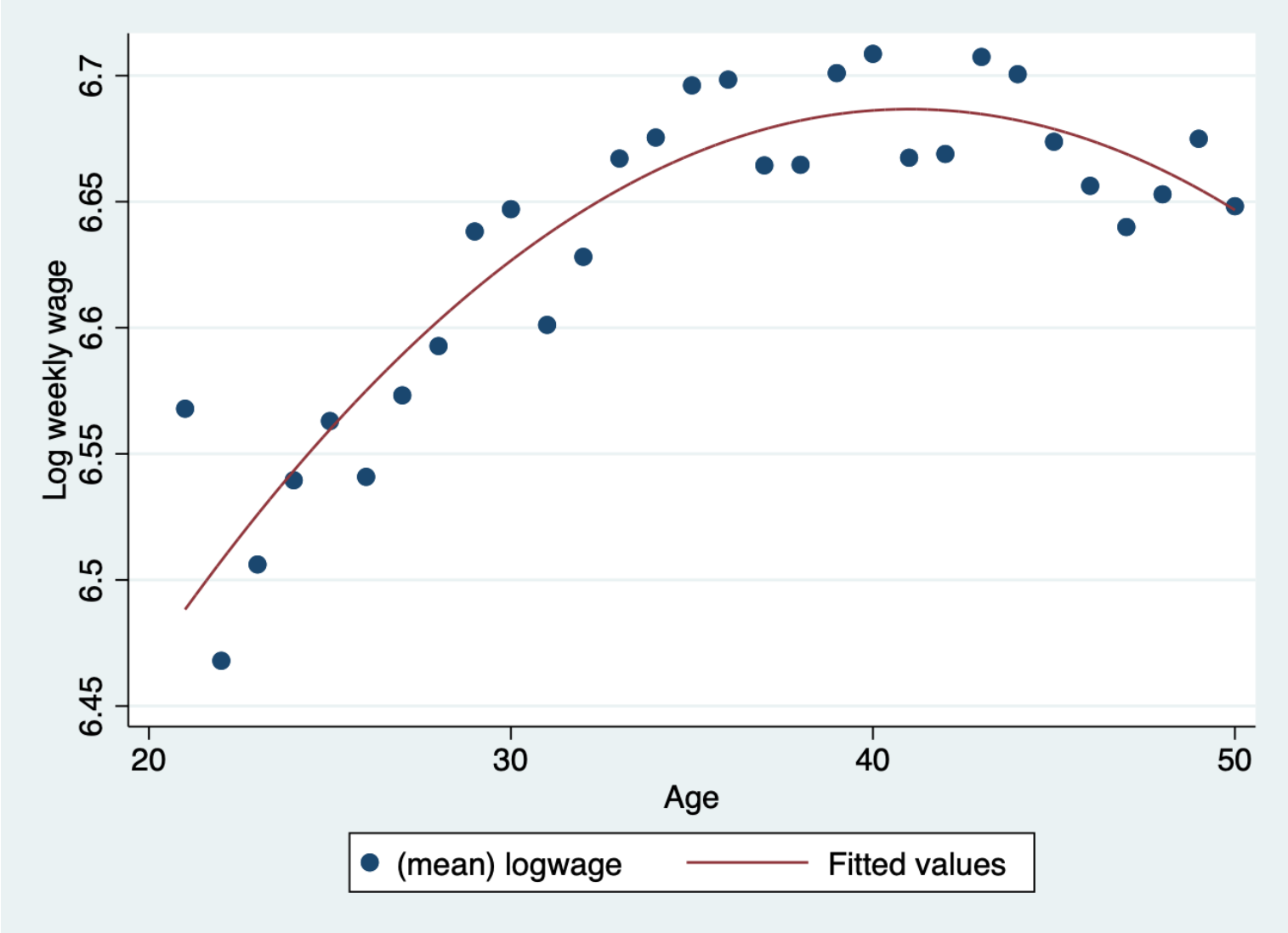
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- We thus now have a general way of estimating $E[Y_i \mid \mathbf{X}_i] \approx \mathbf{X}_i' \boldsymbol{\beta}$ for any vector $\mathbf{X}_i = (1, X_{i1}, \ldots, X_{iK})'$

Quadratic Regression of Log Wages on Age



Constant	5.8591
Age	0.0403
Age^2	-0.0005

Interpreting Quadratic Regression Coefficients

- With a quadratic fit, we have

$$E[Y_i|X_i = x] \approx \hat{\boldsymbol{\beta}}_0 + \hat{\boldsymbol{\beta}}_1 x + \hat{\boldsymbol{\beta}}_2 x^2$$

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- The estimated derivative from a multivariate regression, in this case $\hat{\boldsymbol{\beta}}_1 + 2\hat{\boldsymbol{\beta}}_2 x$, is sometimes called the "marginal effect" at x
  - This terminology is a bit unfortunate: this need not be a *causal* effect, just an estimated derivative of the CEF

Interpreting Our Example Coefficients

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$$\hat{\boldsymbol{\beta}} = \left(\frac{1}{N} \sum_{i=1}^{N} \boldsymbol{X}_i \boldsymbol{X}_i' \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^{N} \boldsymbol{X}_i Y_i \right)$$

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- Using this notation, one can show that $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1} \boldsymbol{X}'\boldsymbol{Y}$

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- **Consistency:** $\hat{\boldsymbol{\beta}} \to_p \boldsymbol{\beta}$.

Asymptotic Properties of Multivariate OLS

- **Asymptotic normality**:

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \to_d \mathrm{N}(0, \boldsymbol{\Sigma}),$$

where $\boldsymbol{\Sigma} = E[\boldsymbol{X}_i \boldsymbol{X}_i']^{-1} Var(\boldsymbol{X}_i \varepsilon_i) E[\boldsymbol{X}_i \boldsymbol{X}_i']^{-1}$ and $\varepsilon_i = Y_i - \boldsymbol{X}_i' \boldsymbol{\beta}$

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where $\hat{\varepsilon}_i = Y_i - \boldsymbol{X}_i' \hat{\boldsymbol{\beta}}$.

- Note: $\hat{\boldsymbol{\Sigma}}$ is a *matrix*.
  - The standard error for $\hat{\boldsymbol{\beta}}_j$ is $\sqrt{\hat{\boldsymbol{\Sigma}}_{jj}}/\sqrt{N}$
  - The off-diagonal elements correspond with covariances between $\hat{\boldsymbol{\beta}}_j, \hat{\boldsymbol{\beta}}_k$

Example - Log Earnings by Age

Variable	Coefficient	SE
Constant (β_0)	5.8591	0.1409
Age (β_1)	0.0403	0.0077
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- Example: we have data from Texas on each county's presidential vote over three elections (2012,2016,2020)

- Let $Y_i =$ Biden vote share in county i, $X_{i1} =$ Clinton vote share in county i, and $X_{i2} =$ Obama vote share in county i

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- This says that

$$E\left[\text{Biden vote} \middle| \text{Clinton vote}, \text{Obama vote}\right] \approx \beta_0 + \beta_1 \times \text{Clinton vote} + \beta_2 \times \text{Obama vote}$$

OLS Estimates

Variable	Coefficient	SE
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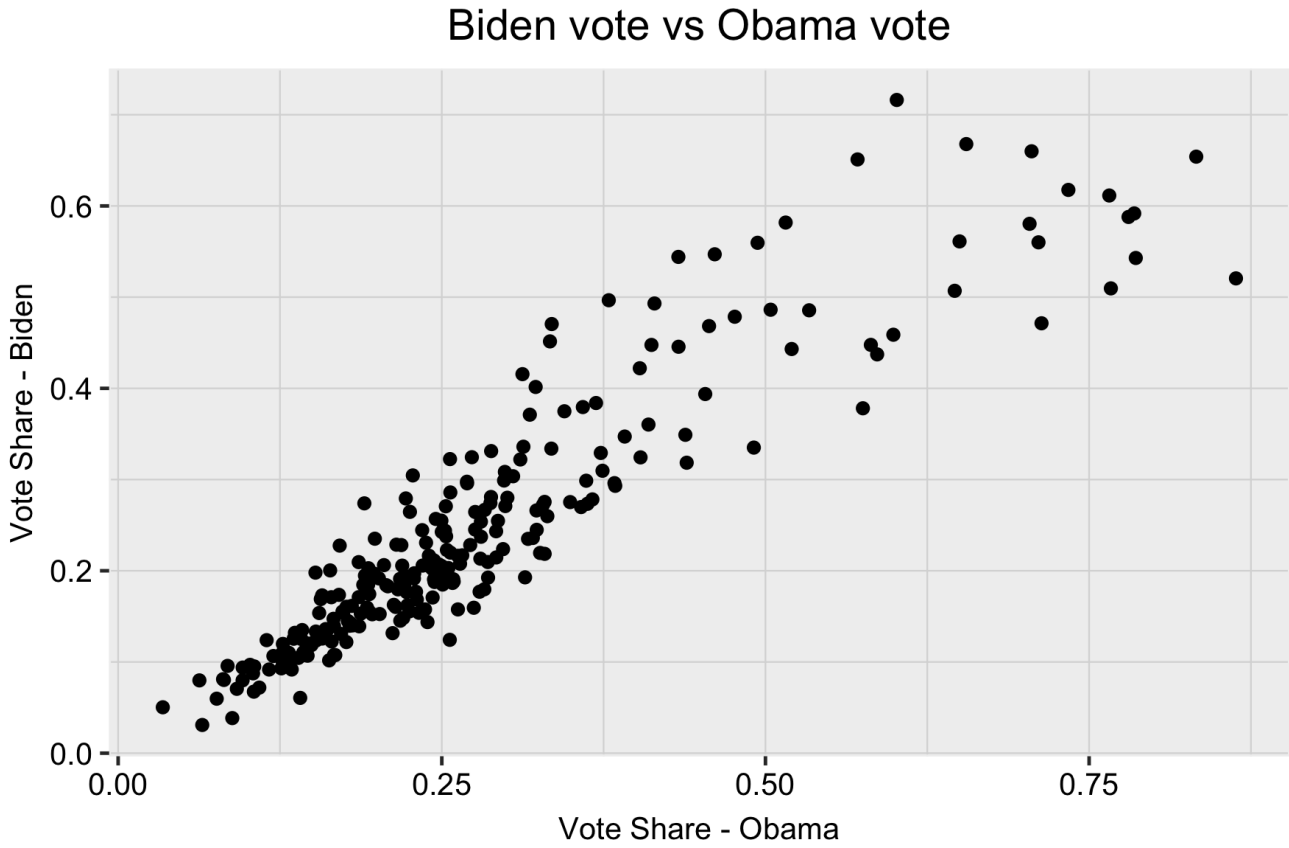
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- Notice that the coefficient on Obama vote share is *negative*
- Wait, does this mean Biden did worse in places that Obama did well?!



- If we look at the data, we see that Biden vote share is highly positively correlated with Obama vote share.
- So what's going on?!

Interpreting Regression Coefficients

- Remember that multivariate OLS is approximating the CEF as

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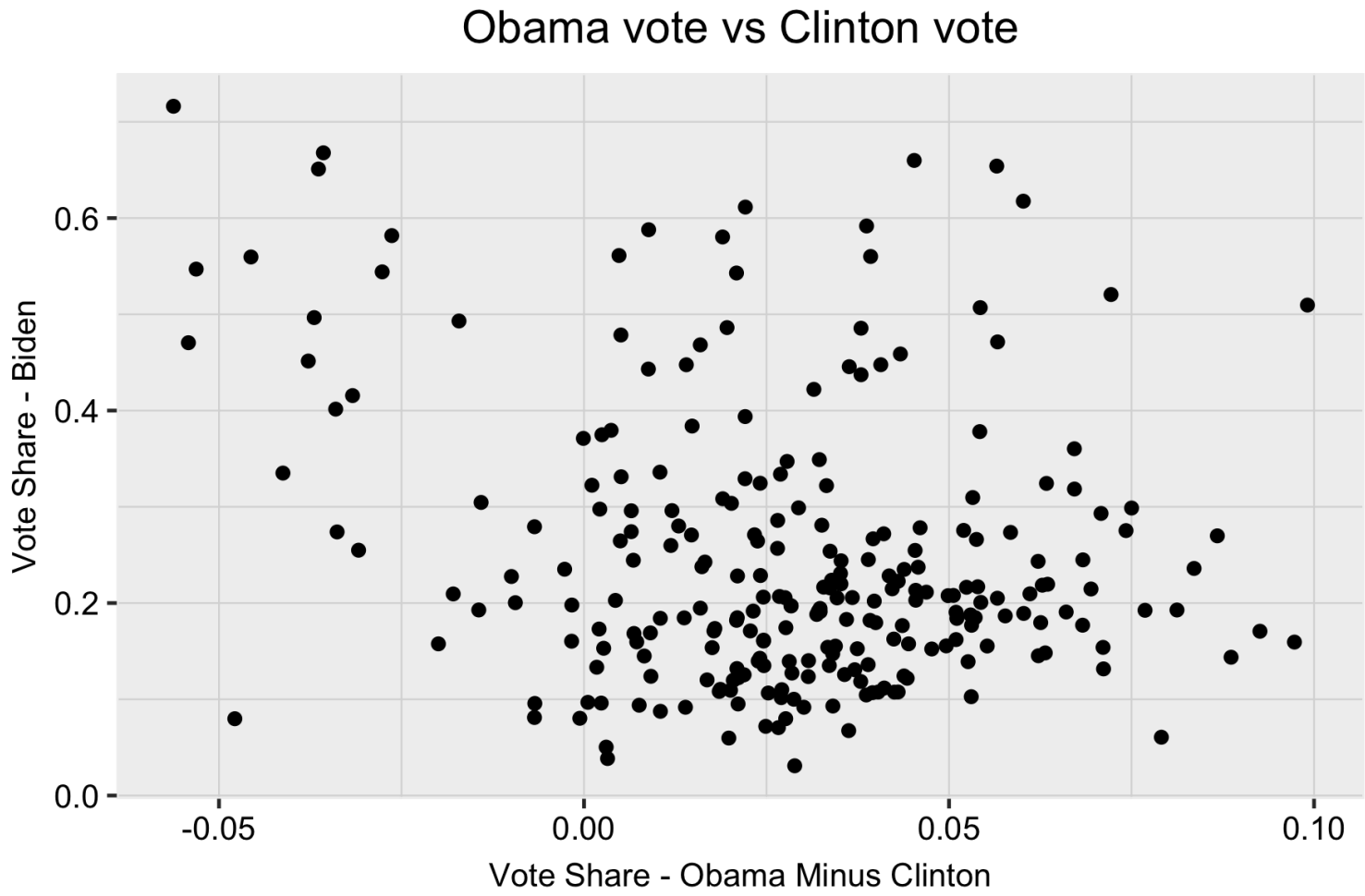
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- In other words, Biden did better in places where Democratic vote share was increasing between 2012 and 2016!


Obama vote vs Clinton vote
Vote Share - Biden
0.0
0.2
0.4
0.6
-0.05
0.00
0.05
0.10
Vote Share - Obama Minus Clinton

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- 3) Obtain $\hat{\beta}_2$ by regressing Y_i on the OLS residual $X_{i2} - \hat{X}_{i2}$:

$$Y_i = \alpha + (X_{i2} - \hat{X}_{i2})\beta_2 + v_i$$

Illustration Using Election Data

- Regress Obama vote share on Clinton vote share

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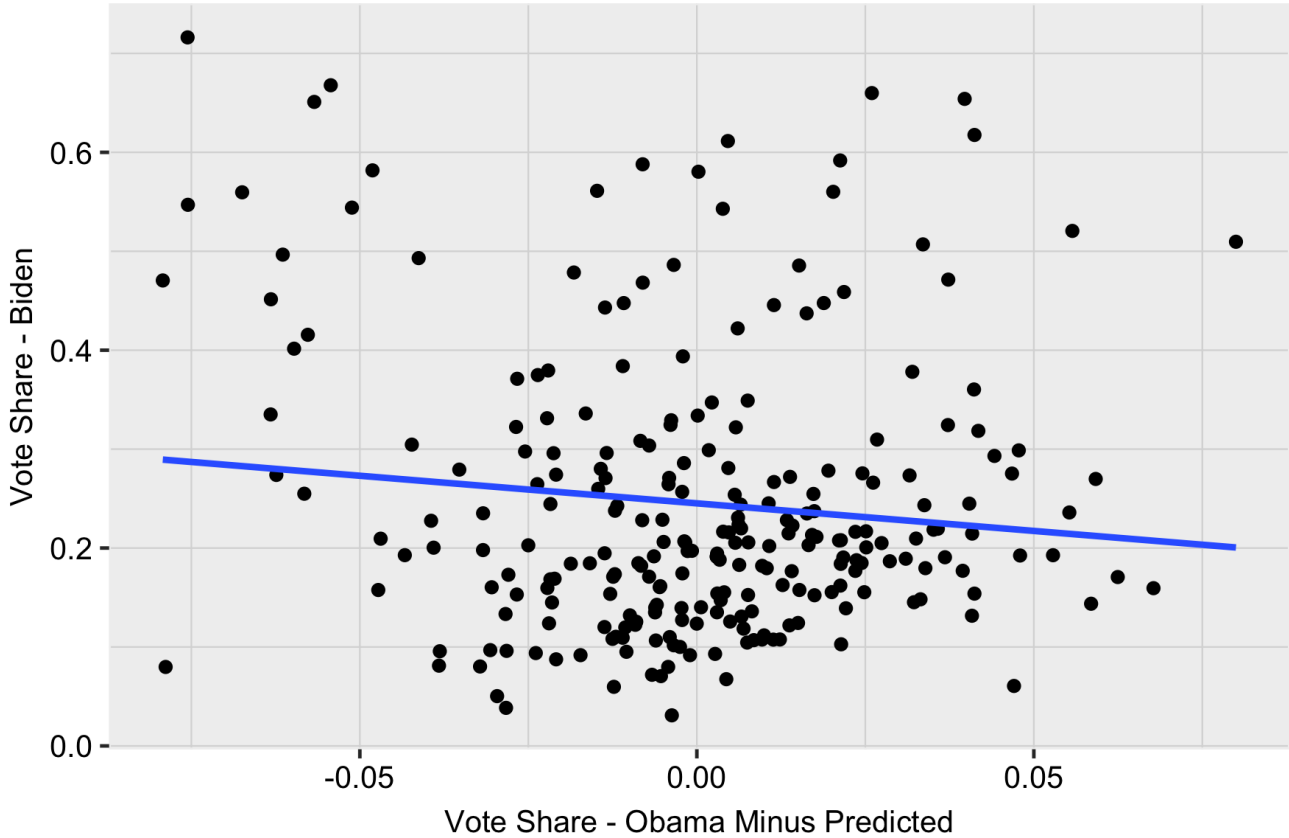
- Predict Obama vote share using Clinton vote share:

$$\hat{X}_{i2} = 0.03 + 0.98 X_{i1}$$

- Regress Biden vote share on $X_{i2} - \hat{X}_{i2}$:

Intercept ($\hat{\alpha}$)	0.25
Obama minus predicted ($\hat{\beta}_2$)	-0.56

- The estimate $\hat{\beta}_2$, -0.56, is exactly what we got before!



- The slope of the best-fit line is precisely $\hat{\beta}_2 = -0.56$.
- FWL generally gives us an easy way to visualize/interpret multivariate regression coefficients

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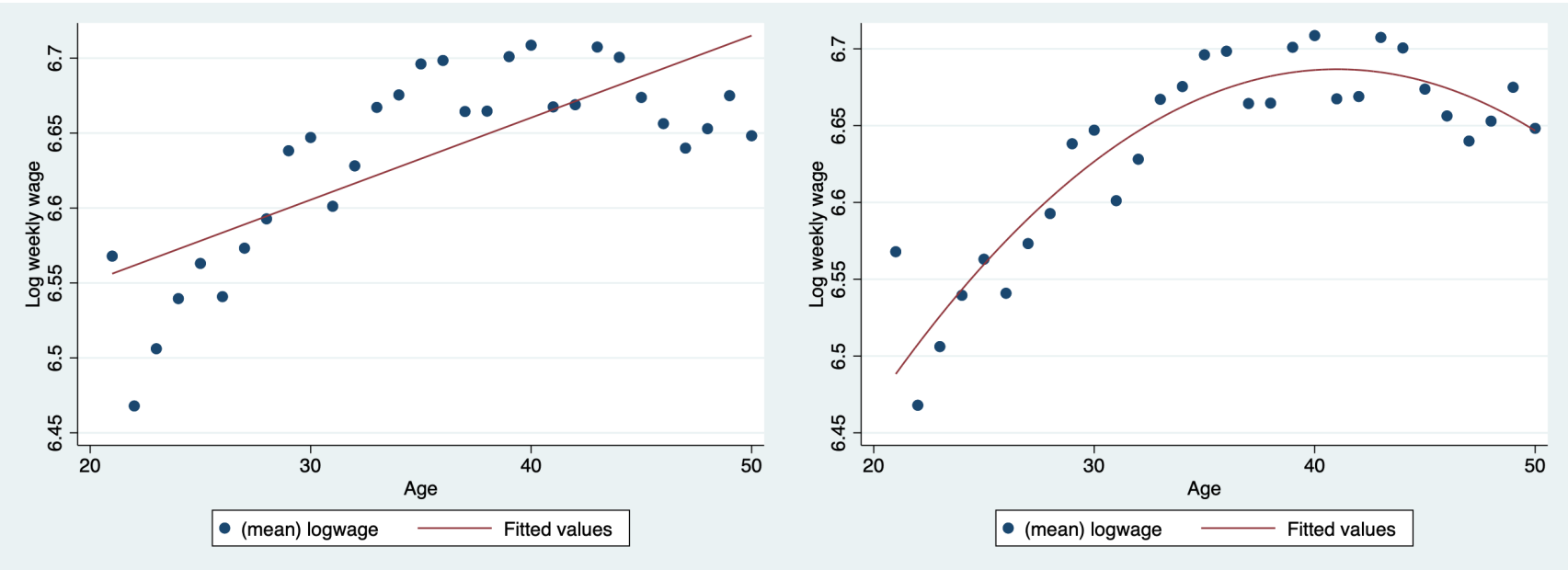
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- To estimate R^2, we replace population values with sample analogs

$$\hat{R}^2 = \frac{\frac{1}{N}\sum_i (\mathbf{X}_i'\hat{\boldsymbol{\beta}} - \bar{\mathbf{X}}_i'\hat{\boldsymbol{\beta}})^2}{\frac{1}{N}\sum_i (Y_i - \bar{Y})^2} = 1 - \frac{\frac{1}{N}\sum_i \hat{\varepsilon}_i^2}{\frac{1}{N}\sum_i (Y_i - \bar{Y})^2}$$

R^2 in the Wage-Age Example



(a) $\hat{R}^2 = 0.44$ (b) $\hat{R}^2 = 0.73$

- The linear fit explains 44% of the variation in average earnings across ages, whereas the quadratic fit explains 73%

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$$\frac{1}{N}\sum_i (\underbrace{Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)}_{\hat{\varepsilon}_{Linear}})^2$$

  While the coefficients in a quadratic fit minimize

$$\frac{1}{N}\sum_i (\underbrace{Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 X_i^2)}_{\hat{\varepsilon}_{Quad}})^2$$

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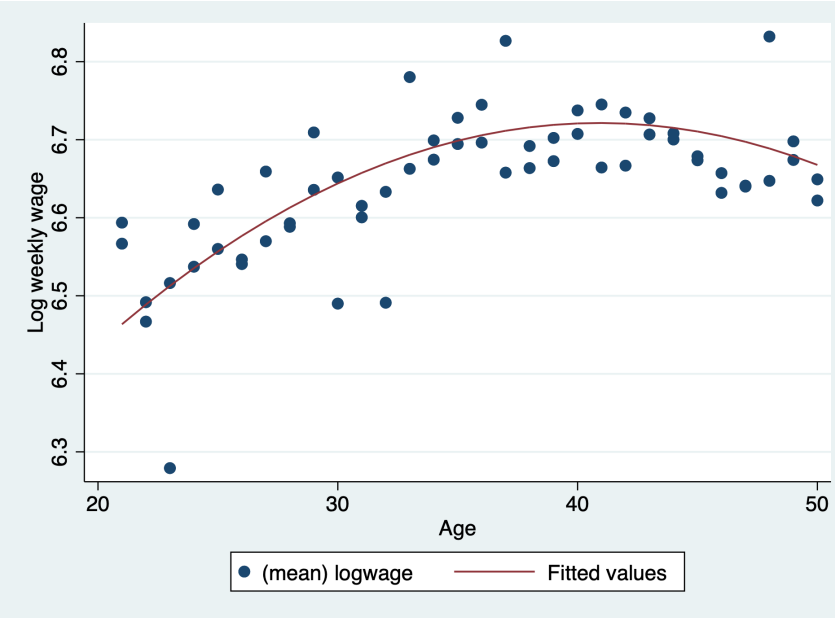
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- But is a more complicated model always better?

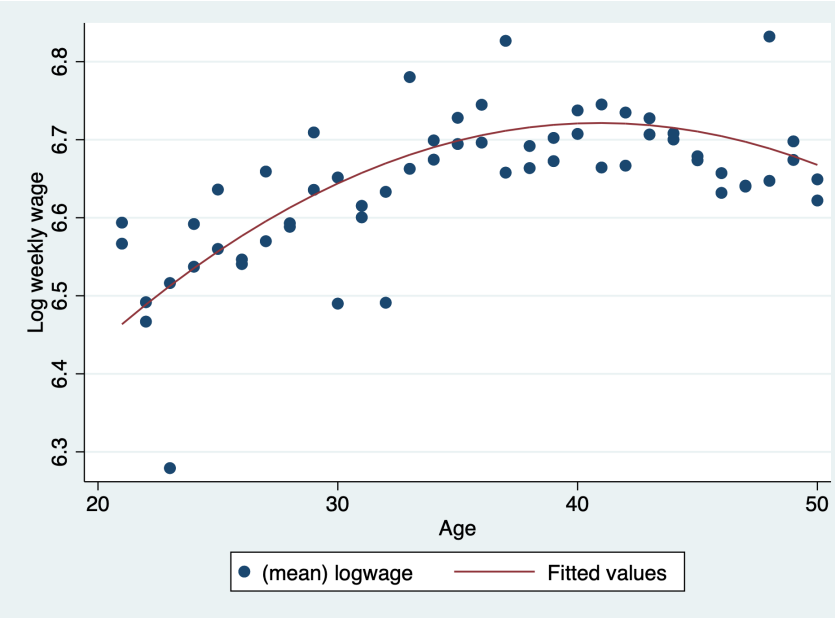


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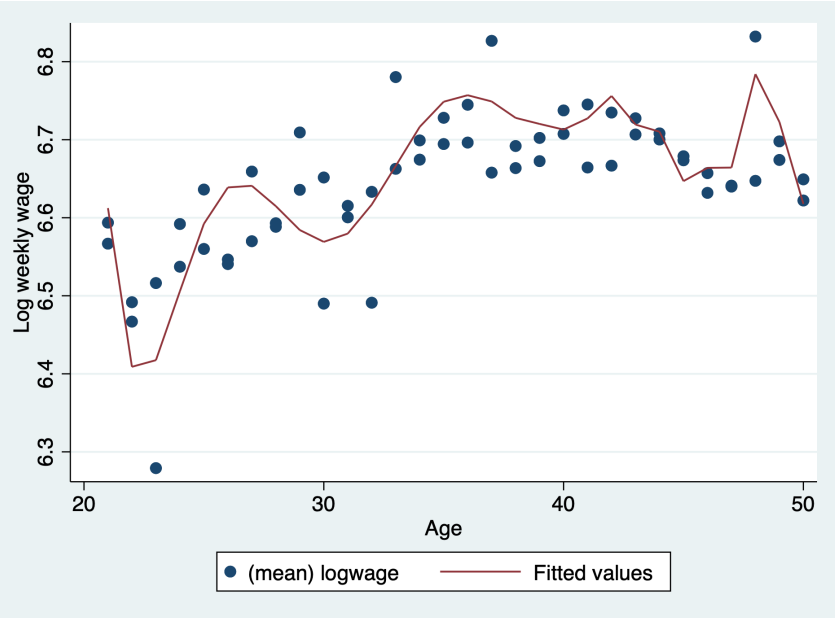


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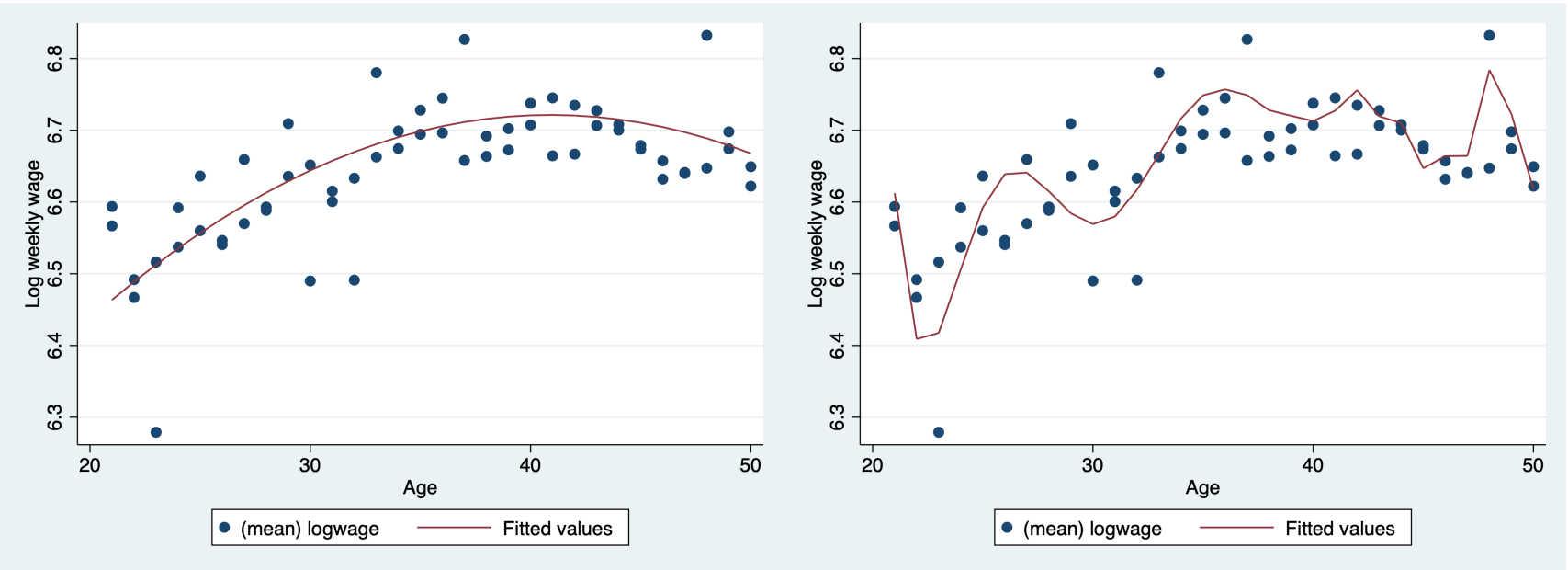


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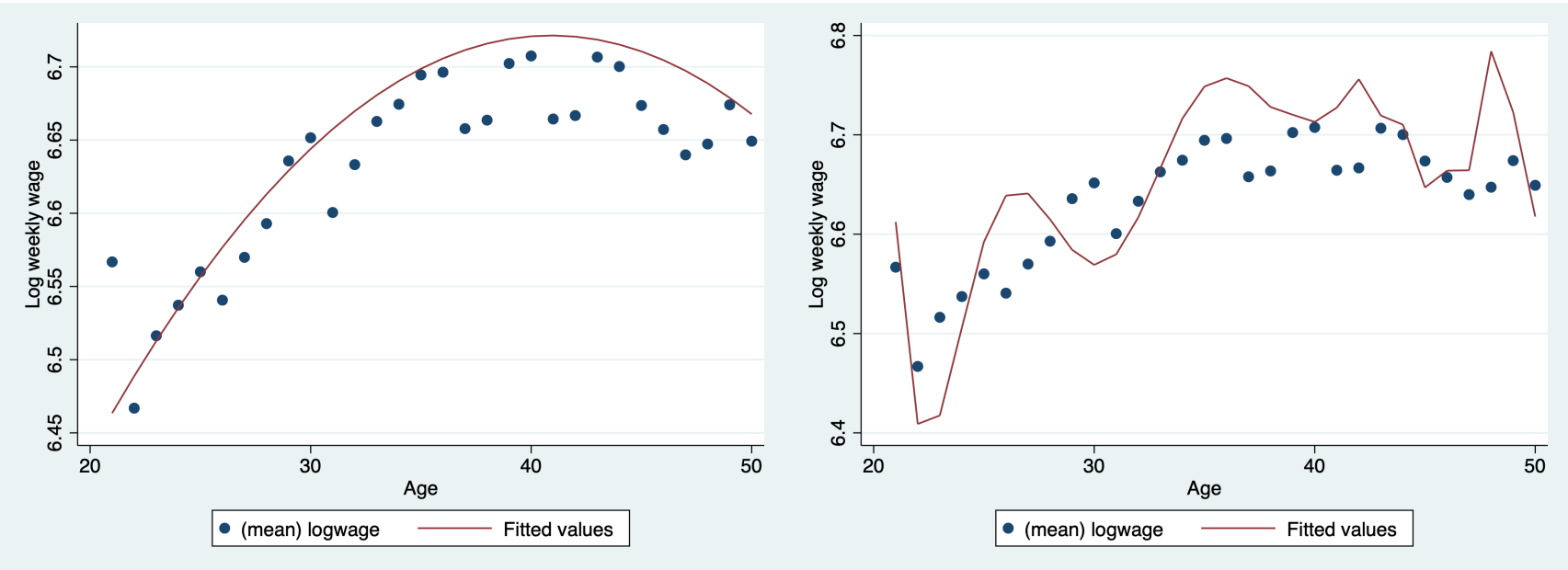
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- No, it looks too "squiggly" – it has adapted to fit the exact points in the sample

- Suppose we draw a new sample and test the prediction of our model trained on the first data-set



(i) Quadratic

(j) 20th order poly

- Suppose we draw a new sample and test the prediction of our model trained on the first data-set
- The quadratic fit generalizes pretty well to the new data.
- But the 20th-order polynomial does very poorly. It "overfit" the features of the specific previous sample. This doesn't generalize well to a new sample

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  - Choose the complexity of the model based on how well it predicts out of sample
  - Cross-validation is the basis of modern *machine learning* (ML) methods. ML is very powerful, and recent work has extended ML for causal problems (but beyond scope of this class)

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- Generally, we will be more confident if the model conclusions are not sensitive to tweaks in the model specification.

Outline

1. Deriving Multivariate Regression and OLS✓

2. Regression and Causality

3. Regression Odds and Ends

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- So if we estimate the multivariate regression

$$Y_i = D_i\beta + \boldsymbol{X}_i'\boldsymbol{\gamma} + \varepsilon_i,$$

  we can interpret $\hat{\boldsymbol{\beta}}$ as an estimate of the ATE.

Dale and Krueger

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- The C&B survey covers students who attended 30 colleges for the high school class of 1978; it contains important variables:
- **Earnings** in 1996
- **College application and demographic variables** including SAT scores, class rank, family income, race, etc
- **College application decisions** — i.e. the set of schools students applied to and were admitted

Institution	School-average SAT score in 1978
Barnard College	1210
Bryn Mawr College	1370
Columbia University	1330
Denison University	1020
Duke University	1226
Emory University	1150
Georgetown University	1225
Hamilton College	1246
Kenyon College	1155
Miami University (Ohio)	1073
Northwestern University	1240
Oberlin College	1227
Pennsylvania State University	1038
Princeton University	1308
Rice University	1316
Smith College	1210
Stanford University	1270
Swarthmore College	1340
Tufts University	1200
Tulane University	1080
University of Michigan (Ann Arbor)	1110
University of North Carolina (Chapel Hill)	1080
University of Notre Dame	1200
University of Pennsylvania	1280
Vanderbilt University	1162
Washington University	1180
Wellesley College	1220
Wesleyan University	1260
Williams College	1255
Yale University	1360

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- If conditional unconfoundedness holds & the regression approx. to the CEF is decent, then β should (approximately) equal the treatment effect of attending a college with higher average SAT scores.

First Pass: SAT Scores and Demographics in X_i

	Full sample
Variable	1
School-average SAT score/100	0.076
	(0.016)
Predicted log(parental income)	0.187
	(0.024)
Own SAT score/100	0.018
	(0.006)
Female	−0.403
	(0.015)
Black	−0.023
	(0.035)
Hispanic	0.015
	(0.052)
Asian	0.173
	(0.036)
Other/missing race	−0.188
	(0.119)
High school top 10 percent	0.061
	(0.018)
High school rank missing	0.001
	(0.024)
Athlete	0.102
	(0.025)

- $\hat{\beta} = 0.076$ indicates about an increase in log wages of 7.6 from attending a school with 100 higher SAT points

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- To address this concern, Dale and Kruger have a second analysis: control for the set of colleges that a student applied to / was admitted
  - Are able to see this because of the C&B data

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- The variables $X_{i1}, ..., X_{i4}$ are often called "fixed effects" for the set of schools you were admitted to.

Introduction to "Fixed Effects"

- We can then approximate the CEF as

$$E[Y_i|\boldsymbol{X}_i, D_i] = D_i\beta_D + X_{i1}\beta_1 + X_{i2}\beta_2 + X_{i3}\beta_3 + X_{i4}\beta_4$$

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- Intuitively, β_D represents the average difference from going to an elite school *among* students who got into the same set of schools

Applicant Group	Student	Private			Public			1996 Earnings
		School I	School II	School III	School IV	School V	School VI	
A	1		Reject	**Admit**		Admit		110,000
	2		Reject	**Admit**		Admit		100,000
	3		Reject	Admit		**Admit**		110,000
B	4	**Admit**			Admit		Admit	60,000
	5	Admit			Admit		**Admit**	30,000
C	6		**Admit**					115,000
	7		**Admit**					75,000
D	8	Reject			**Admit**	Admit		90,000
	9	Reject			Admit	**Admit**		60,000

- "Applicant groups" all applied + admitted to the same set of schools
- The school a student actually attended is highlighted

Controlling for Application Decisions in Practice

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- As an approximation, they group colleges into bins based on average-SAT rounded to nearest 25 points
- They then control for the set of colleges you applied/were admitted to based on these bins (e.g., X_{i1} might correspond to being rejected at a school w/SAT 1350 and accepted at 2 schools w/SAT 1250)

Variable	Basic model: no selection controls		Matched-applicant model
	Full sample	Restricted sample	Similar school-SAT matches*
	1	2	3
School-average SAT score/100	0.076	0.082	−0.016
	(0.016)	(0.014)	(0.022)
Predicted log(parental income)	0.187	0.190	0.163
	(0.024)	(0.033)	(0.033)
Own SAT score/100	0.018	0.006	−0.011
	(0.006)	(0.007)	(0.007)
Female	−0.403	−0.410	−0.395
	(0.015)	(0.018)	(0.024)
Black	−0.023	−0.026	−0.057
	(0.035)	(0.053)	(0.053)
Hispanic	0.015	0.070	0.020
	(0.052)	(0.076)	(0.099)
Asian	0.173	0.245	0.241
	(0.036)	(0.054)	(0.064)
Other/missing race	−0.188	−0.048	0.060
	(0.119)	(0.143)	(0.180)
High school top 10 percent	0.061	0.091	0.079
	(0.018)	(0.022)	(0.026)
High school rank missing	0.001	0.040	0.016
	(0.024)	(0.026)	(0.038)
Athlete	0.102	0.088	0.104
	(0.025)	(0.030)	(0.039)

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- This indicates about a -1.6 log wage return to attending a school with 100 higher SAT points (but not significant!)

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  - Students may choose to go to a lower-ranked school only if it has a particularly good program in what they're interested in
- It's also important to realize that the schools in the C&B study tend to be selective. These results can at best be interpreted as the causal effect between attending a selective school and highly-selective school

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- How will the coefficients we estimate be biased if we forget to include some variables?
- To answer this question, we will derive what is called the **omitted variable bias** (OVB) formula

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$$\tilde{\beta}_D = \frac{Cov(Y_i, D_i)}{Var(D_i)} = \frac{Cov(\beta_0 + \beta_D D_i + \beta_1 X_{i1} + e_i, D_i)}{Var(D_i)}$$

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&= \beta_D + \beta_1 \frac{Cov(X_{i1}, D_i)}{Var(D_i)}
\end{aligned}$$

where we use $E[e_i] = E[D_i e_i] = 0$ from the FOCs for regression

Evaluating OVB

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- Hence $\tilde{\beta}_D$ will be very biased for β_D if the omitted variable X_{i1} is both highly correlated with Y_i and highly correlated with D_i
  - On the flip side, if either $\beta_1 = 0$ or $\gamma_D =$ then we have no OVB!

OVB Formula in Finite Samples

- We just showed that the coefficients from the population regressions

$$Y_i = \beta_0 + \beta_D D_i + \beta_1 X_i + e_i \tag{1}$$

$$Y_i = \tilde{\beta}_0 + \tilde{\beta}_D D_i + \varepsilon_i \tag{2}$$

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- It turns out that the OLS estimates for these two regressions have the same relationship (just replace pop means w/sample means in all formulas)

$$\hat{\tilde{\beta}}_D = \hat{\beta}_D + \hat{\beta}_1 \frac{\widehat{Cov}(X_{i1}, D_i)}{\widehat{Var}(D_i)}$$

OVB Illustration

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- Let's think about what would happen if we forgot to control for SAT

- Here are the results that A&P get when controlling for SAT:

Variable	Coefficient	SE
Private school ($\hat{\beta}_D$)	0.095	0.052
SAT score /100 ($\hat{\beta}_1$)	0.048	0.009
Constant	[...]	[...]

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Private school ($\hat{\gamma}_D$)	.83
Constant	[...]

- Here are the results that A&P get when controlling for SAT:

Variable	Coefficient	SE
Private school ($\hat{\beta}_D$)	0.095	0.052
SAT score /100 ($\hat{\beta}_1$)	0.048	0.009
Constant	[...]	[...]

- What would happen if we omitted the control for SAT score from the regression? How would $\hat{\beta}_D$ change?

- To compute this, we need to know the coefficients from the regression of X_{i1} (SAT score/100) on D_i (Private school):

Variable	Coefficient
Private school ($\hat{\gamma}_D$)	.83
Constant	[...]

- Thus, if we omitted X_{i1} our estimated coefficient on private school would be $\hat{\gamma}_D \times \hat{\beta}_1 = 0.83 \times 0.048 \approx 0.04$ larger.

- Indeed, if we actually run the regression omitting SAT scores, we see that the coefficient on private school is 0.04 larger.

- Results including SAT score:

Variable	Coefficient	SE
Private school (β_D)	0.095	0.052
SAT score /100 (β_1)	0.048	0.009
Constant	[...]	[...]

- Results excluding SAT score:

Variable	Coefficient	SE
Private school ($\tilde{\beta}_D$)	0.135	0.055
Constant	[...]	[...]

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  - If SAT score were more strongly related to earnings ($\hat{\beta}_1$ larger)
  - If treatment were more strongly correlated with SAT scores ($\hat{\gamma}_D$ larger)

Omitted Variable Bias Formula - Multiple Variables

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- How does $\tilde{\beta}_D$ relate to β_D? Answer:

$$\tilde{\beta}_D = \beta_D + \beta_2 \gamma_D$$

where γ_D is the coefficient from the regression

$$X_{i2} = \gamma_0 + \gamma_D D_i + \gamma_1 X_{i1} + u_i$$

- This is similar to the OVB formula we had from before, except everything controls for X_{i1}

Evaluating the Bias (Again)

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  - OVB is positive if and only if the correlations are the same sign

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- Let's think about what happens if we forgot the control for SAT score

- Here are the results that A&P get when controlling for both SAT score and set of schools you're admitted to:

Variable	Coefficient	SE
Private school ($\hat{\beta}_D$)	0.003	0.039
SAT score /100 ($\hat{\beta}_2$)	0.033	0.007
College admitted ($\hat{\boldsymbol{\beta}}_1$)	[...]	[...]

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- If we are interested in heterogeneity by X_{i1}, we can estimate:

$$Y_i = \beta_D D_i + \beta_{DX}(D_i \times X_{i1}) + \boldsymbol{\gamma}'\boldsymbol{X}_i + \varepsilon_i$$

Dale and Krueger - Heterogeneous Effects

- DK estimate a regression of the form:

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where D_i is average-SAT score/100, X_{i1} is log family income, and $\boldsymbol{X}_i$ includes indicators for your attended college + other controls

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$$\hat{\beta}_D + \hat{\beta}_{DX} \times ln(100,000)$$

OF MALE AND FEMALE FULL-TIME WORKERS

	Parameter estimates	
	Basic model: no selection controls	Matched-applicant model*
Variable	1	2
School-average SAT score/100	0.701	0.537
	(0.185)	(0.224)
Predicted log(parental income)	0.915	0.819
	(0.212)	(0.247)
Predicted log of parental income * school SAT score/100	−0.063	−0.056
	(0.019)	(0.023)
Own SAT score/100	0.018	−0.011
	(0.006)	(0.007)
High school top 10 percent	0.062	0.080
	(0.019)	(0.026)
High school rank missing	0.005	0.018
	(0.024)	(0.038)
Athlete	0.104	0.105
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- In DK, students at the 10th percentile of family earnings have log parental income of 8.86 ($\approx \$7,000$)
- What is the estimated effect of going to a school w/avg SAT 100 points higher for students at the 10th percentile?

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OF MALE AND FEMALE FULL-TIME WORKERS

Variable	Parameter estimates: Basic model: no selection controls (1)	Parameter estimates: Matched-applicant model* (2)
School-average SAT score/100	0.701	0.537
	(0.185)	(0.224)
Predicted log(parental income)	0.915	0.819
	(0.212)	(0.247)
Predicted log of parental income * school SAT score/100	−0.063	−0.056
	(0.019)	(0.023)
Own SAT score/100	0.018	−0.011
	(0.006)	(0.007)
High school top 10 percent	0.062	0.080
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- What might explain why elite college seems to matter more for students from poor backgrounds?
- Not entirely clear... networking more important for students who don't have as many family connections?

Linear Combinations of Coefficients

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- Suppose we want to construct a CI or test hypothese about $CATE(x)$. How can we do that?

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- So we can form a CI for $\beta_1 + \beta_2 x$ using $\hat{\beta}_1 + \hat{\beta}_2 x \pm 1.96\hat{\sigma}_x/\sqrt{N}$, where $\hat{\sigma}_x^2 = \hat{\Sigma}_{11} + x^2 \hat{\Sigma}_{22} + 2x\hat{\Sigma}_{12}$.

Example

- Recall that in DK we have $\hat{\beta}_D = 0.537$ and $\hat{\beta}_{DX} = -0.056$.
- Suppose that the part $\hat{\mathbf{\Sigma}}/N$ corresponding w/the coefficients of interest is

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lincom

- The `lincom` command in Stata generalizes the argument above to test for any linear combination of coefficients, i.e. parameters of the form $a_1\beta_1 + ... + a_k\beta_k$

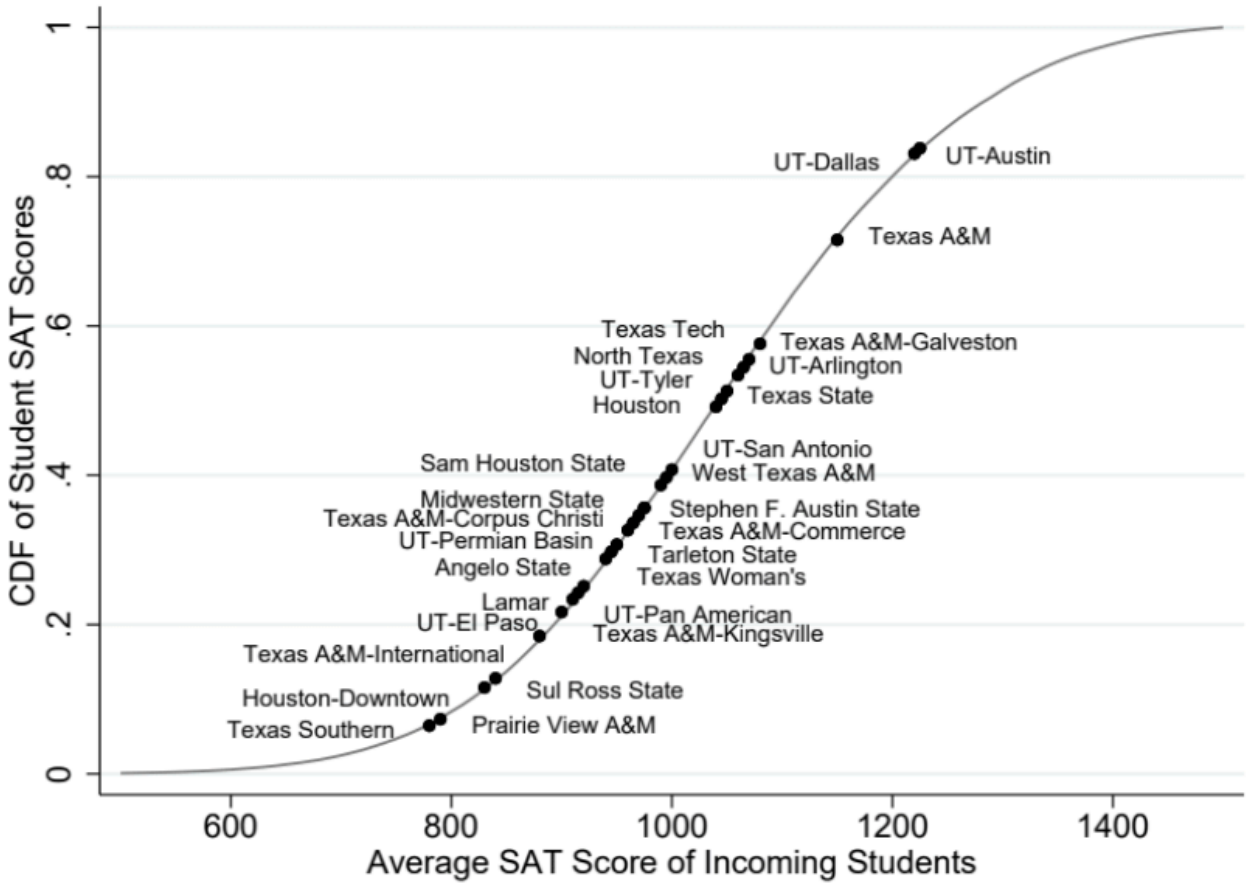
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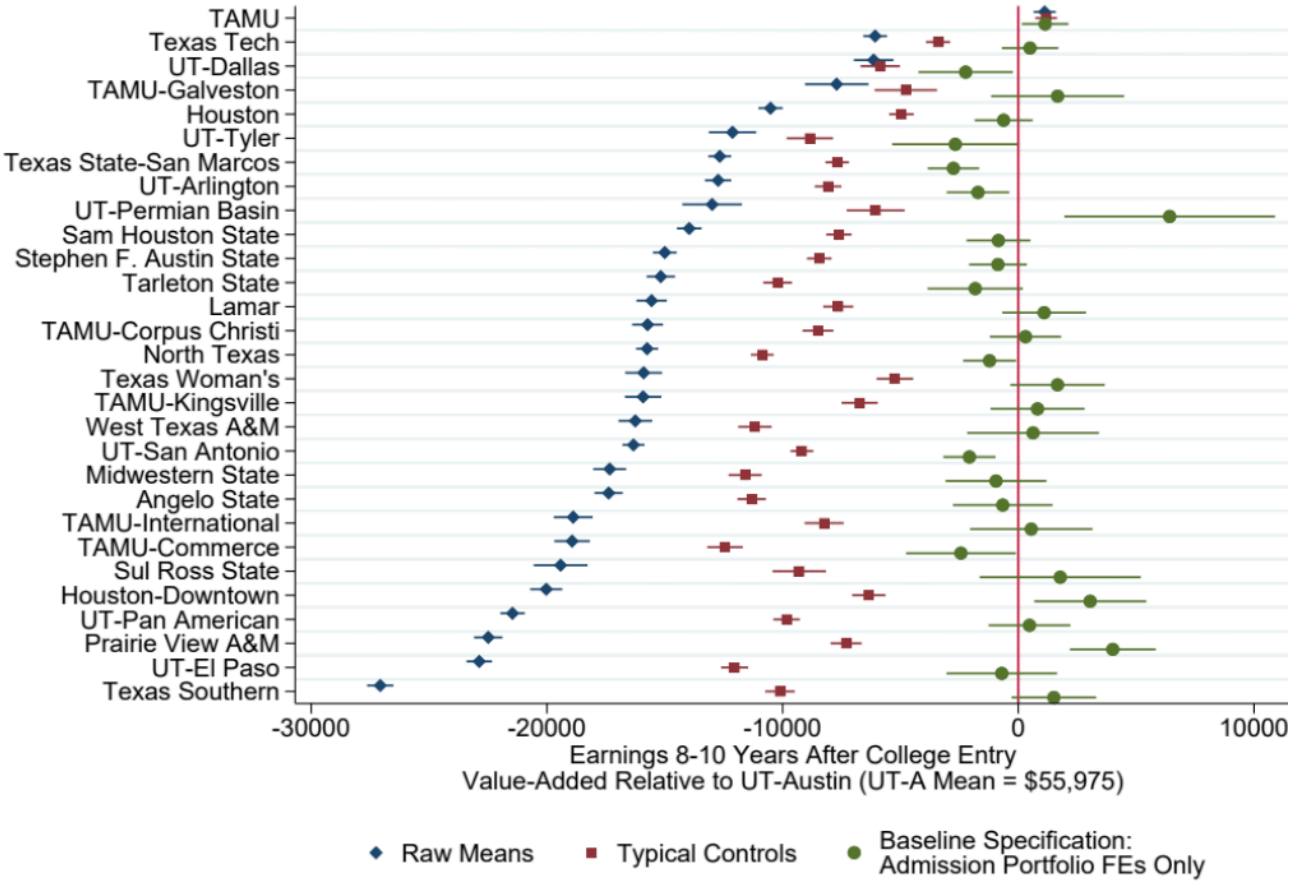
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- Similar (but slightly more complicated) asymptotic arguments can be used to test hypotheses on non-linear combinations of coefficients, e.g. $\beta_1\beta_2 + \beta_3^2$. This can be done using the `nlcom` command in Stata.

More Evidence - Mountjoy and Hickman (2021)

- Mountjoy and Hickman do a modern version of Dale and Krueger using data on all 4-year public universities in Texas
- Sample includes all students graduating from HS in TX from 1999-2008
- Very large sample allows them to control for exact set of colleges applied/admitted to, and to estimate effects for each university (422K students attend 4-year college; 126K admitted to multiple schools w/matches)
- Only 55% enroll in the most selective school they were admitted to

Figure 1: Selectivity Across Colleges and SAT Score Variation Within Colleges

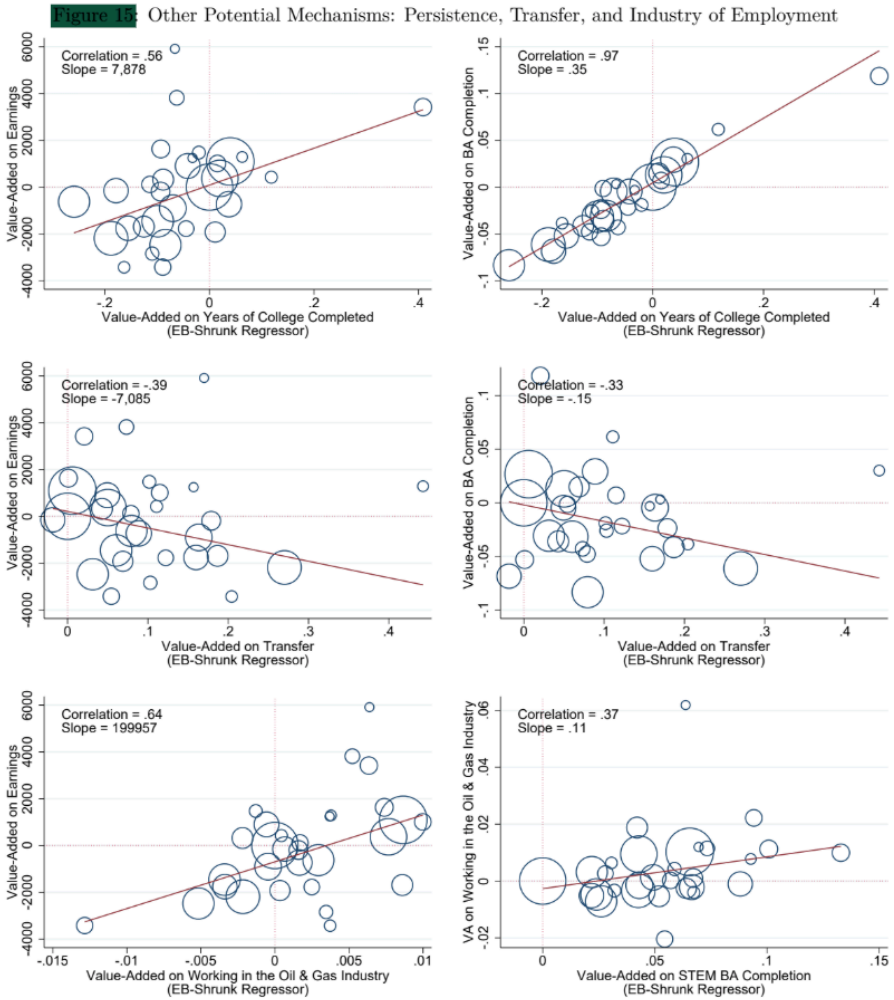




- There are some significant earnings diffs across colleges, but relatively small
- More prestige does not necessarily mean higher earnings (UT-Austin vs Permian Basin)

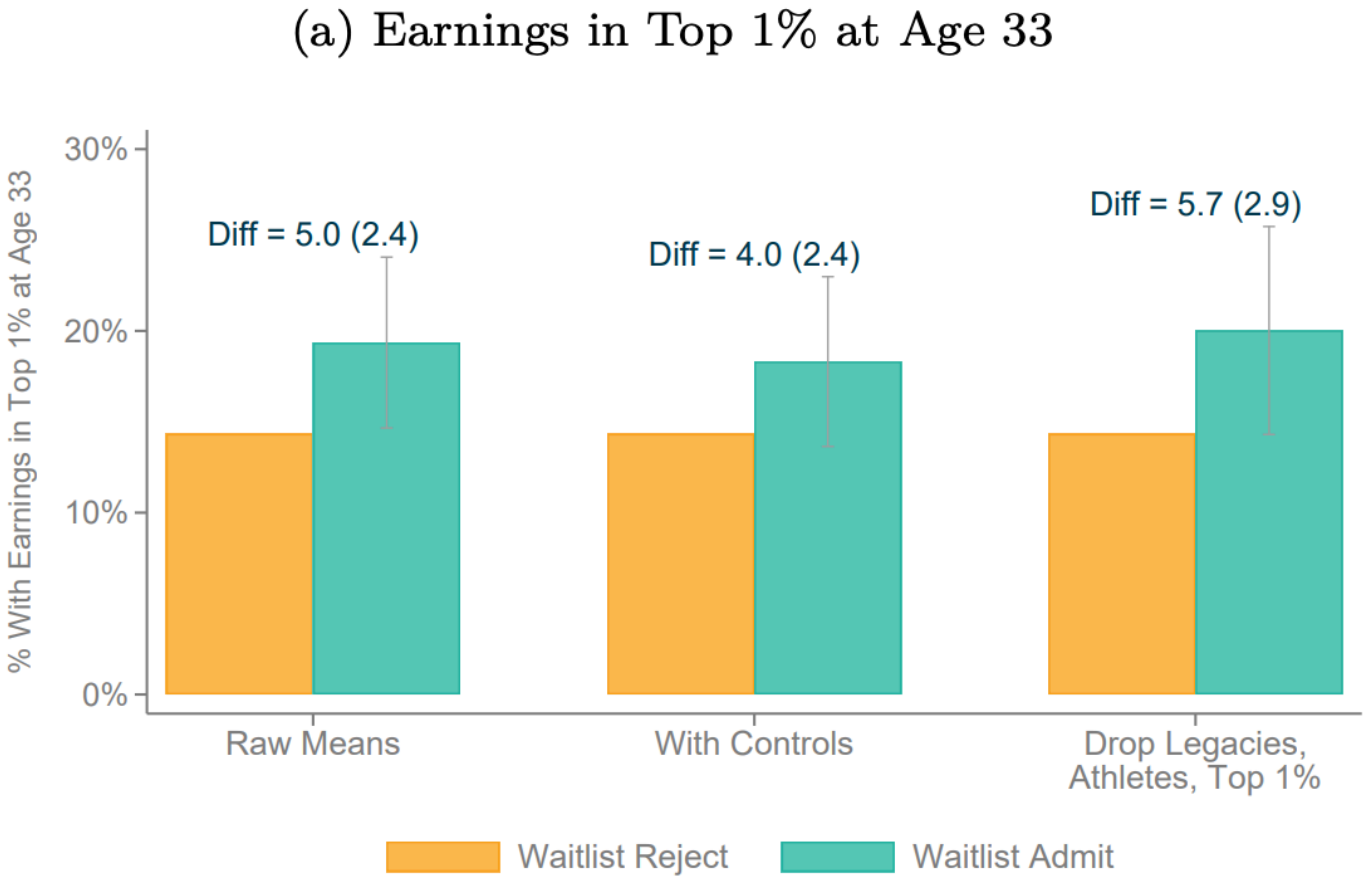
What are the higher value-added colleges doing

Higher college completion, higher STEM completion, higher employment in oil & gas

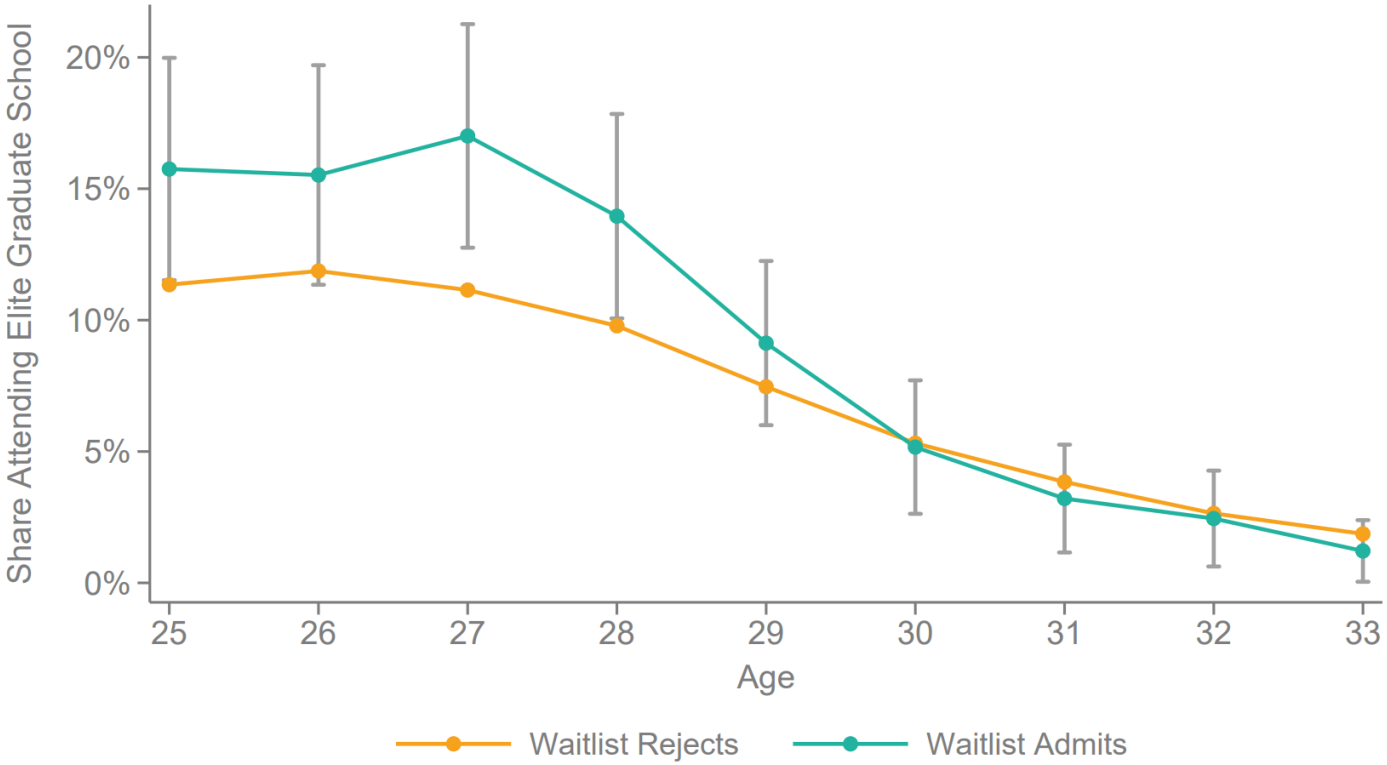


Chetty, Deming, Friedman (2025)

- Combine admissions data from several Ivy+ schools (Ivies, Stanford, MIT, Chicago, Duke) w/tax data on earnings
- Compare students on the waitlist to Ivy+ that get admitted versus not
    - Admission off waitlist at one Ivy+ doesn't predict outcome at others
- Like Dale and Krueger, they don't find much effect on *average* earnings
- But they do find impacts on "elite" outcomes — earnings in top 1%, working at "elite firm", attending "elite" gradaute school


(a) Earnings in Top 1% at Age 33
% With Earnings in Top 1% at Age 33
30%
20%
10%
0%
Diff = 5.0 (2.4)
Diff = 4.0 (2.4)
Diff = 5.7 (2.9)
Raw Means
With Controls
Drop Legacies, Athletes, Top 1%
Waitlist Reject
Waitlist Admit

(b) Elite Graduate School Attendance



Outline

1. Deriving Multivariate Regression and OLS✓

2. Regression and Causality✓

3. Regression Odds and Ends

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  - Are the effects bigger for male athletes from rich families than female non-athletes from poor families?
- There are a large number of hypotheses that we may want to test!
- This can lead to what's called a **multiple hypothesis testing** problem

Multiple Hypothesis Testing (Cont.)

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- Thus, if we test for a significant effect among the entire population, if there is truly no effect we should reject the null only 5% of the time.
- But suppose we first test for a significant effect among men. And we then also test for a significant effect among women.
- If there is no significant effect among either group, what is the probability that we find at least one significant effect?

- Suppose the samples for men and women are drawn independently.
- Let N_{sig} be the number of significant results.

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- So we'll find at least one significant effect almost 10% of the time if there is no effect for either group
- By the same argument, if we test the null for K independent groups, and there is no true effect, we will reject the null with probability $1 - 0.95^K$

Probability of finding at least one subgroup with a significant effect with K independent groups (and zero treatment effect):

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2	0.0975
3	0.1426
5	0.2262
10	0.4013
100	0.9941

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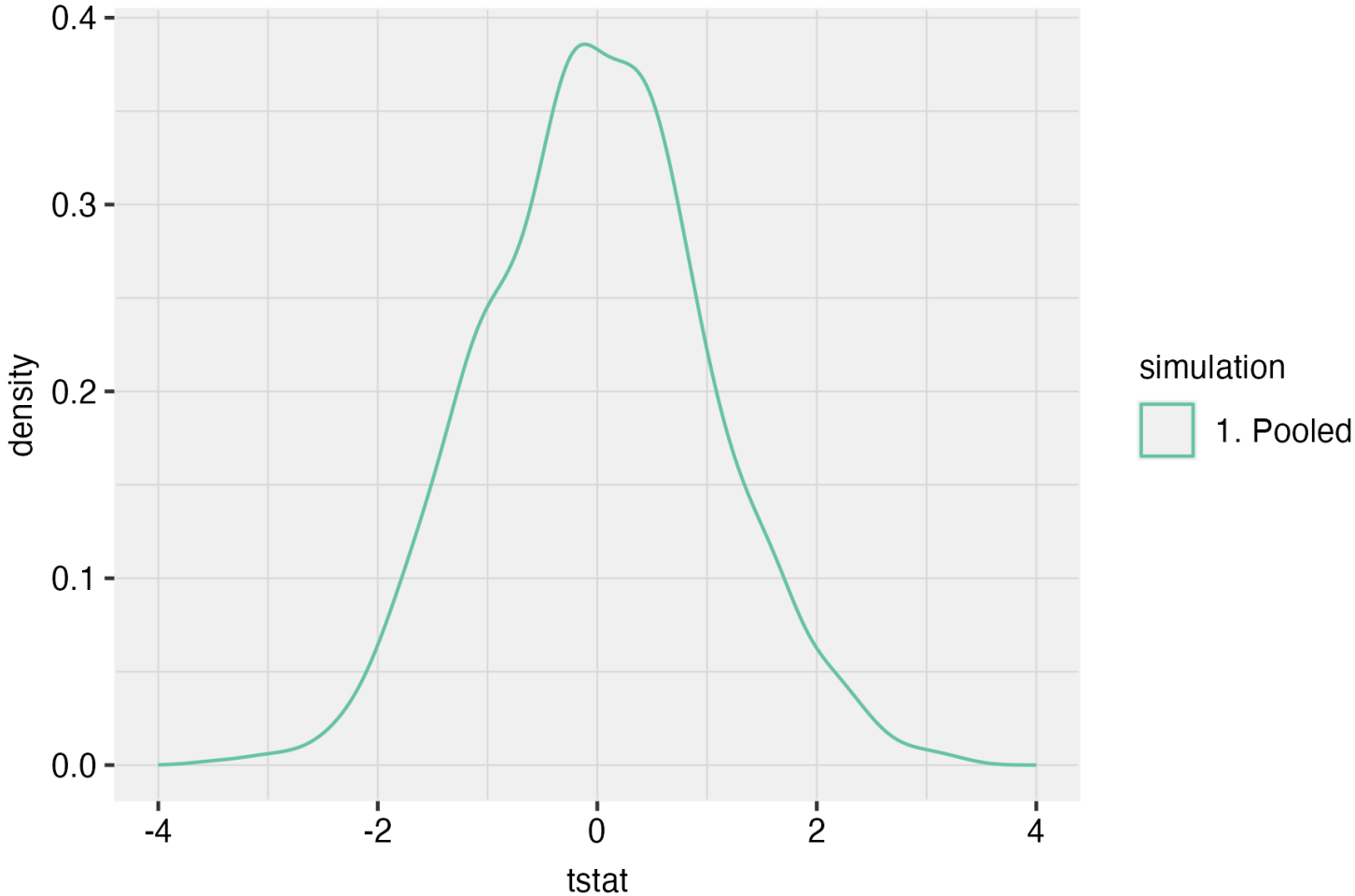
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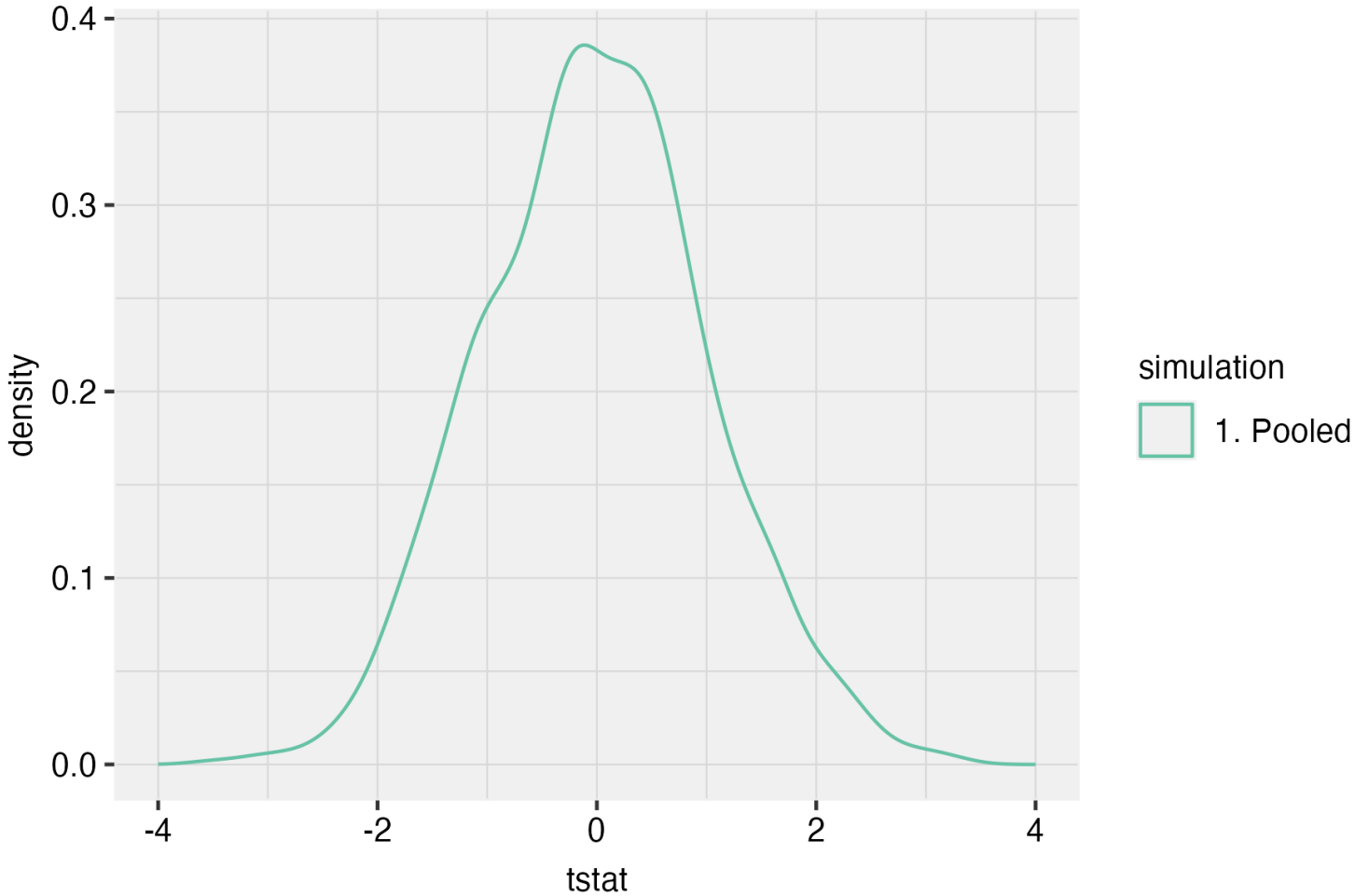
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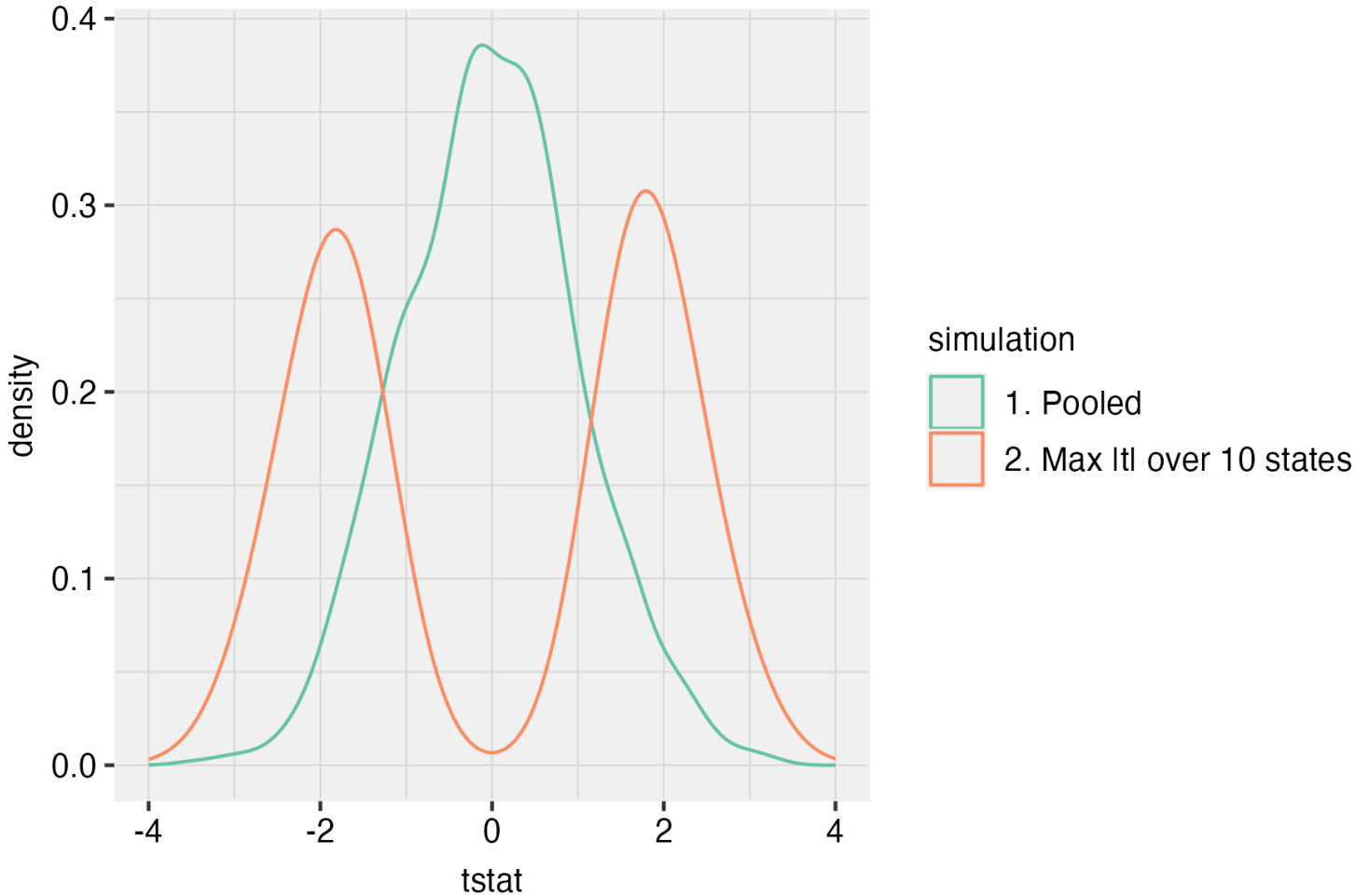
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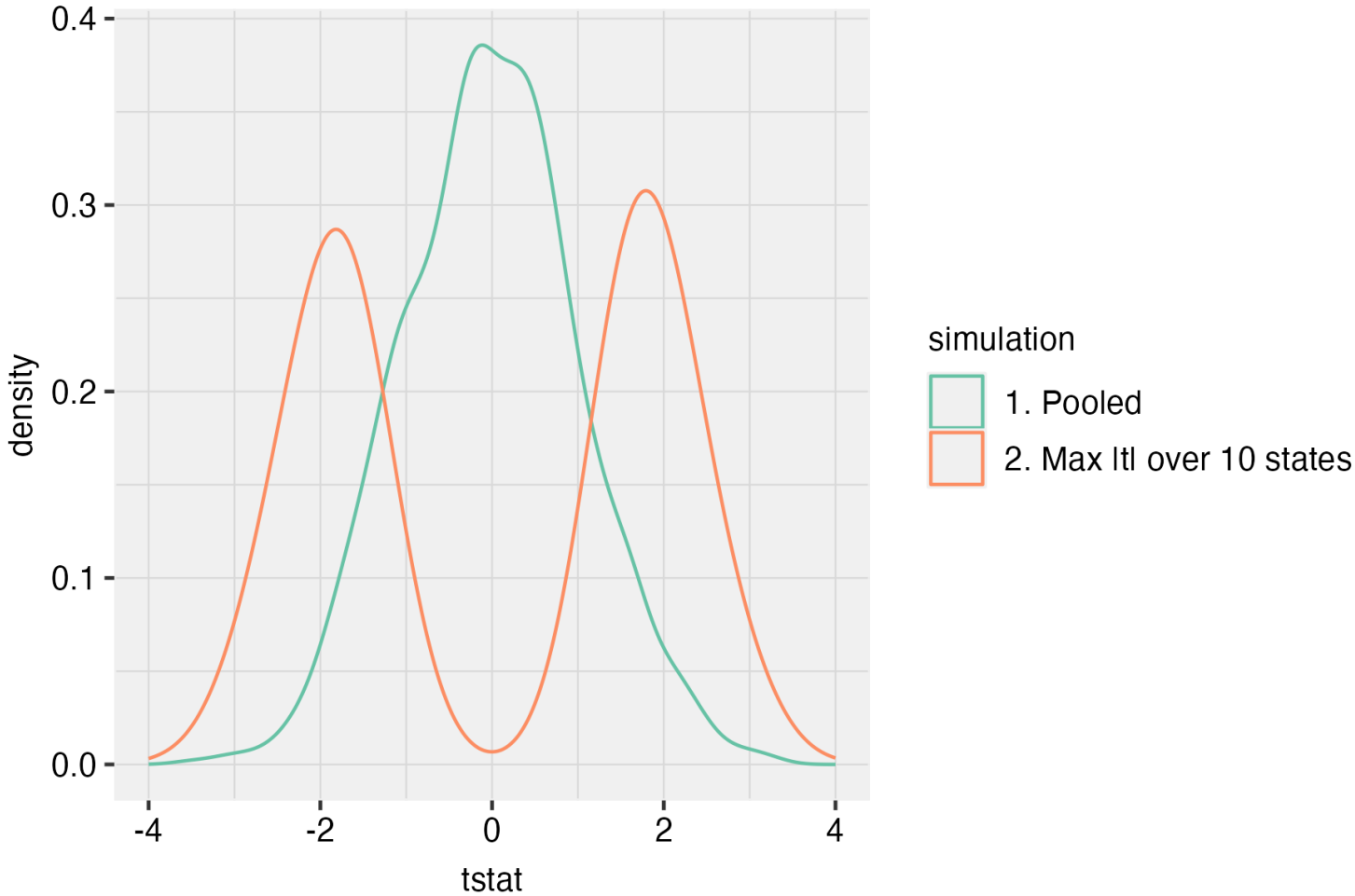
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- I then calculate the fraction of simulations in which we get at least one significant estimate


0.4
0.3
0.2
0.1
0.0
density
-4
-2
0
2
4
tstat
simulation
1. Pooled

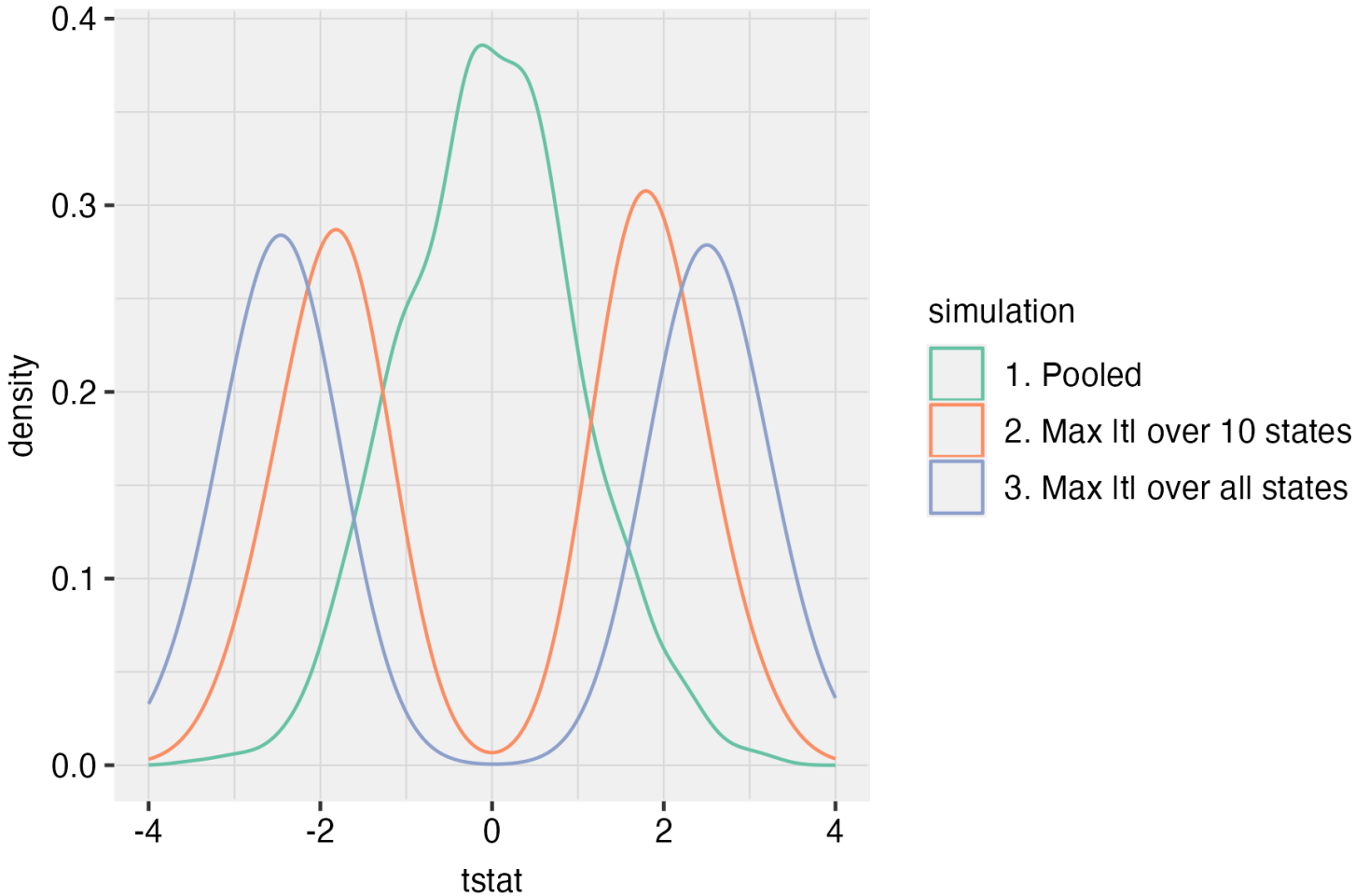


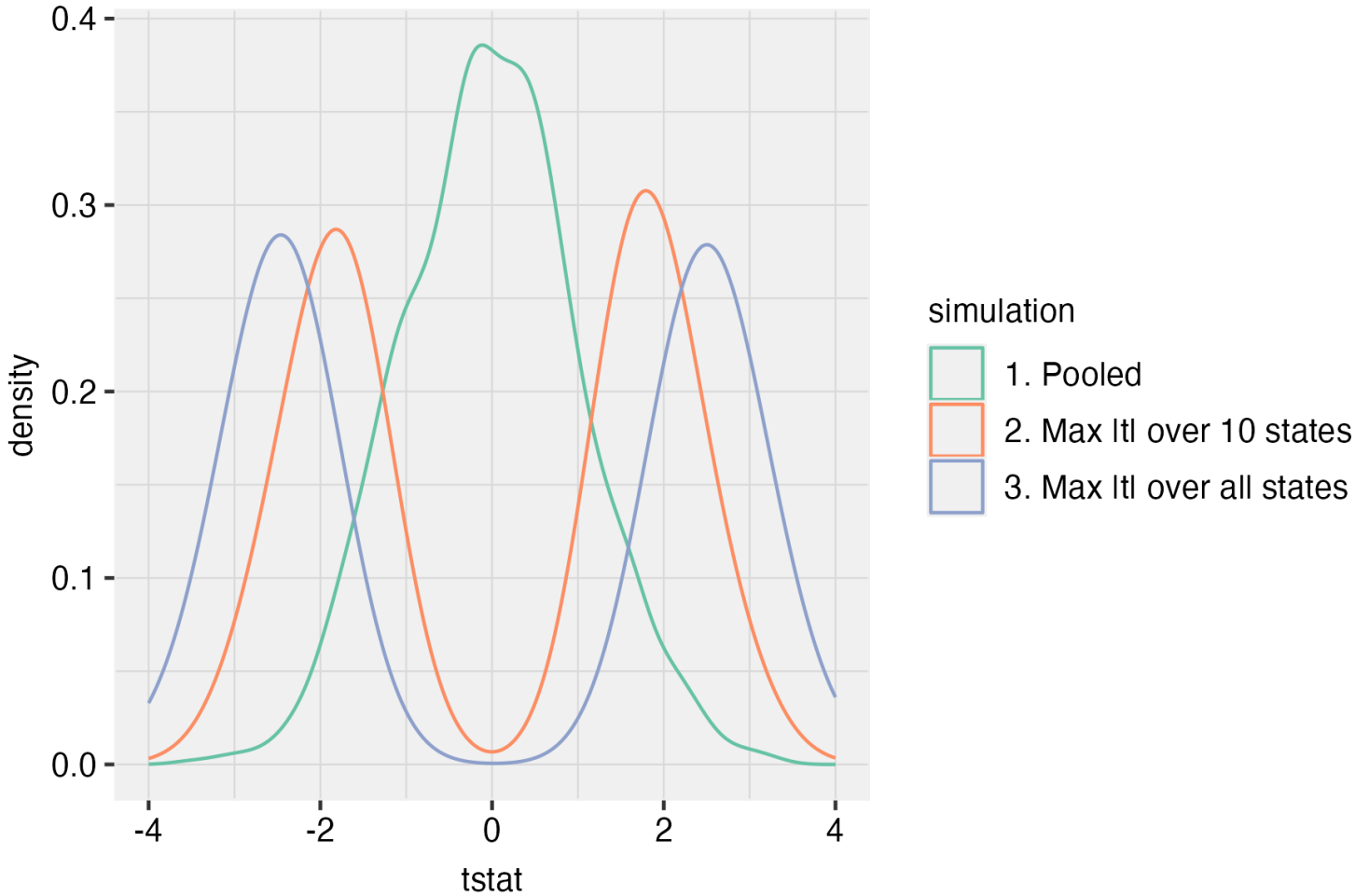
- When testing the pooled effect across all states, we find a significant effect 6% of the time


0.4
0.3
0.2
0.1
0.0
density
-4
-2
0
2
4
tstat
simulation
1. Pooled
2. Max |t| over 10 states



- When testing the pooled effect across all states, we find a significant effect 41% of the time


0.4
0.3
0.2
0.1
0.0
density
-4
-2
0
2
4
tstat
simulation
1. Pooled
2. Max |t| over 10 states
3. Max |t| over all states



- When testing the pooled effect across all states, we find a significant effect 93% of the time

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- Downside: if you have lots of hypotheses, power against any one hypothesis can be low (e.g., 99.9% confidence intervals are very wide). And the test is generally conservative in the sense that we find any significant effect $< 5\%$ of the time

Probability of rejecting at least one hypothesis without and with Bonferroni correction

Simulation	Uncorrected	Corrected
Pooled	0.06	0.05
10 States	0.41	0.05
50 States	0.93	0.04

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- How do we do this?

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- Consider the *F-statistic*: $F = \sum_k \hat{t}_k^2$. By the continuous mapping theorem, if all K null hypotheses are satisifed

$$F \to_d \sum_k Z_k^2 \text{ for } Z \sim N(0, \mathbf{I})$$

  This is called a *chi-squared* (χ^2) distribution with K degrees of freedom

- We can therefore test that all K null hypotheses are satisfied by comparing F to the 95th percentile of the chi-squared dist (say c). If $F > c$, we reject that all hypotheses are valid.

- This approach is called an *F-test*.

Probability of rejecting the joint null that the treatment effect is zero for all states

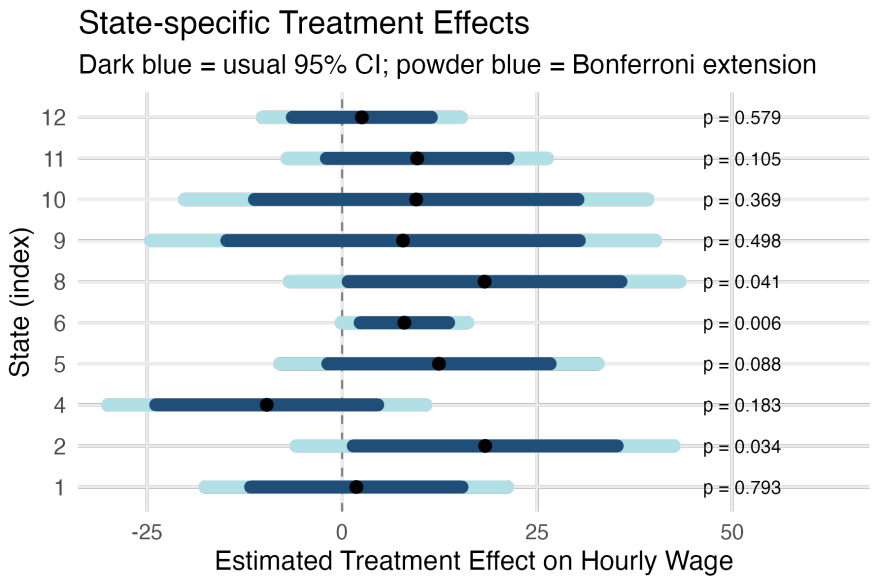
Simulation	Individual tests	Joint test
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F-tests can have better power than Bonferroni

- Consider modified simulation design where we add \$6 to wages if you get our fake treatment
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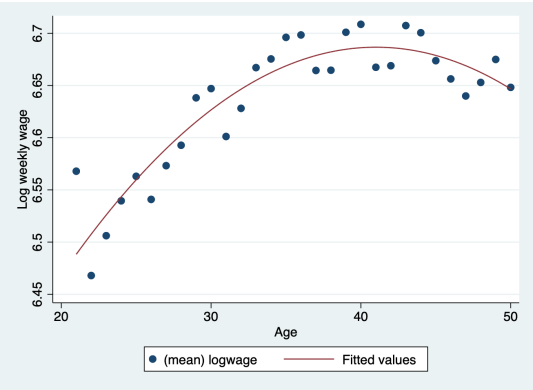
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- Here's an example from one simulation



No state-level effects are significant after Bonf adjustment. But p-value for joint test is 0.004.

Testing multiple regression coefficients

- Sometimes we may want to test the null hypothesis that multiple regression coefficients are equal to zero (e.g. both linear and quadratic terms are zero in our earnings vs age example)



- We can also use an F-test for this problem, we just need to account for the correlation in the coefficients
- We showed in earlier classes that $\sqrt{N}(\hat{\beta} - \beta_0) \rightarrow_d N(0, \mathbf{\Sigma})$. To test the null $H_0 : \beta = \beta_0$, we then use $F = \sqrt{N}\hat{\mathbf{\Sigma}}^{-\frac{1}{2}}(\hat{\beta} - \beta_0)$
- This is implemented with the `test` function in Stata. E.g., `test age agesq` tests the null that the coefficients on age and age-squared are both zero (so wages don't depend on age)

```
. reg logwage age agesq, r

Linear regression                               Number of obs     =         30
                                                F(2, 27)          =      30.81
                                                Prob > F          =     0.0000
                                                R-squared         =     0.8412
                                                Root MSE          =     .02622

------------------------------------------------------------------------------
             |               Robust
     logwage |  Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0403444   .0076871     5.25   0.000     .0245718     .056117
       agesq |   -.000492   .0001011    -4.87   0.000    -.0006994   -.0002846
       _cons |   5.859108   .1409008    41.58   0.000     5.570003    6.148212
------------------------------------------------------------------------------

.
. test age agesq

 ( 1)  age = 0
 ( 2)  agesq = 0

       F(  2,    27) =   30.81
            Prob > F =    0.0000
```